Viraj Induruwa

virajinduruwa123@gmail.com +94 75 541 5575 https://www.virajinduruwa.com/
https://www.linkedin.com/in/viraj-induruwa/ https://github.com/Viraj-005
https://medium.com/@virajinduruwa2

PROFILE SUMMARY

Backend and Full-Stack Software Engineer with hands-on experience designing and deploying scalable RESTful APIs using FastAPI, Flask, and Django, backed by strong database engineering skills in PostgreSQL and Oracle Database. Proven track record of building and deploying production systems that automated business workflows, reduced manual processing time by up to 90%, and served enterprise users on live infrastructure. Experienced with AI and ML integration including LLM pipelines and RAG systems, authentication and access control (JWT, OAuth, Azure AD), containerized deployments (Docker), and cloud infrastructure (AWS EC2, Lambda, S3, Azure, GCP).

EDUCATION

Horizon Campus Malabe, Sri Lanka
BSc (Hons) in Information Technology May 2021 – Oct 2025
CGPA: 3.30/4.00 | SECOND UPPER CLASS HONOURS
Relevant Coursework: Software Engineering, Operating Systems, Artificial Intelligence, Data Science

SKILLS

- **Programming Languages:** Python, JavaScript, SQL
- **Backend & API Development:** FastAPI, Flask, Django, RESTful API Design, JWT, OAuth, Microservices, System Design, Authentication & Authorization
- **Databases & Data Engineering:** PostgreSQL, Oracle Database, MySQL, SQLite, Database Schema Design, Pandas, NumPy, ETL Concepts
- **Frontend:** React, Tailwind CSS
- **Cloud, DevOps & Tools:** AWS (EC2, Lambda, S3, RDS), Azure Active Directory, Google Cloud Platform (GCP), Docker, Linux, Git, GitHub, Bitbucket, Postman, Playwright
- **AI / Machine Learning:** TensorFlow, PyTorch, Keras, Scikit-Learn, Hugging Face Transformers, OpenCV, YOLO, Tesseract OCR, RAG Systems, LLM Integration
- **Professional Skills:** Analytical Thinking, Problem Solving, Attention to Detail, Team Collaboration, Communication, Leadership, Adaptability, Project Management, Time Management

WORK EXPERIENCE

Freelance Software Engineer - Sri Lanka Telecom (SLT) Oct 2025 – Present

- Continued development and enhancement of the **Sales Incentive Automation System** on SLT's internal infrastructure, deployed on **Linux servers** with **Azure AD** authentication and **Oracle DB** integration
- Led the development of a new **Manager Incentive Module**, automating managerial-level incentive calculations with advanced business logic
- Currently developing two additional automated calculation services to extend the incentive automation platform's coverage across further business workflows
- Transitioned business workflows from manual Excel and Tableau-based processing to a centralized automated system, reducing reporting turnaround time from **7 days to under 3 minutes**
- Enhanced operational efficiency by **90%** while significantly reducing human calculation errors

Intern Software Engineer - Sri Lanka Telecom (SLT) Apr 2025 – Oct 2025

- Contributed to the development of the **Sales Incentive Automation System**, replacing manual Excel-based workflows used by sales and dealer management teams
- Designed and implemented a **Python-based console application** to automate complex sales incentive and dealer commission calculations
- Developed backend APIs using **Django and FastAPI**, integrating with **PostgreSQL and Oracle Database**
- Implemented secure authentication and role-based access control using **Azure Active Directory (Azure AD), Google OAuth, and JWT**
- Built **Python-based data processing pipelines** for validation, transformation, and automation of business data used in incentive calculations
- Improved calculation accuracy and reduced manual processing time by approximately **90%**

PROJECTS

LoopLab IMS - Intern Management System (Production)

- Designed and deployed a full-stack Intern Management System in production, with a FastAPI backend on AWS EC2 and a React frontend on AWS Amplify, serving live users at careers.looplab.lk
- Built async backend architecture using SQLAlchemy (async) and asyncpg with PostgreSQL on AWS RDS, including database schema design and Alembic migrations
- Implemented RESTful APIs with structured authentication flows and file storage integration via AWS S3
- Managed end-to-end infrastructure including systemd service deployment, Nginx reverse proxy configuration, and SSL via Certbot
- Tools Used: FastAPI, React, PostgreSQL, AWS EC2, AWS RDS, AWS S3, AWS Amplify, SQLAlchemy, Alembic, Nginx

AI-Powered Hospital Management System

- Developed an AI-powered Hospital Management System integrating predictive analytics for inventory, equipment maintenance, and patient load forecasting
- Built four machine learning engines for drug expiry prediction, inventory consumption forecasting, equipment maintenance analysis, and patient traffic forecasting
- Tools Used: React, FastAPI, PostgreSQL, CatBoost, Scikit-Learn, AWS EC2, Tailwind CSS

Cancer Detective PRO

- Built a Django and React web app for early detection of skin, lung, and leukemia cancers using 30K+ medical images, achieving up to 98% detection accuracy
- Containerized FastAPI-based ML inference APIs with Docker and deployed on AWS Lambda for scalable real-time predictions

Publications

- **A Web Application for Personalized Book Recommendation** *Jan 2025*

  DOI: 10.13140/RG.2.2.36308.72322

- **Design and Implementation of a Web-Based Platform for Medical Image Analysis: Ethical and Regulatory Considerations** *Feb 2025*

  DOI: 10.13140/RG.2.2.25097.92002

REFEREES

Mrs. Thameera Edirisinghe
Manager - Software Development, Digital Platforms
Sri Lanka Telecom PLC
thameera@slt.com.lk